p2p-netcat

Netcat addressed by PeerId

One cryptographic address for terminal, TCP, UDP, and proxy access across NAT.

IDENTITY

PeerId instead of IP

ROUTE

First authenticated route

SESSION

TCP · UDP · PTY · SOCKS

Your home server has no public address — but you still need SSH.

WITHOUT P2P-NETCAT

The network edge is not yours

Laptop

Home router
CGNAT

Port forwarding is unavailable. Dynamic DNS does not help. A hosted VPN adds another control plane.

WITH P2P-NETCAT

The server keeps one identity

12D3KooW…YXdUfbR9

discover · authenticate · connect

Run a listener at home. From anywhere, dial its PeerId and forward local port 2222 to SSH.

A full Internet tunnel should still work when the gateway is behind NAT.

TRAVEL · LAPTOP

WireGuard

Endpoint
127.0.0.1:15182

p2p-nc

keeps carrier
outside wg0

policy route protects the carrier

UNTRUSTED NETWORK

hotel · airport · mobile hotspot
no inbound port required

HOME · GATEWAY

p2p-nc

UDP listener
→ UDP :51820

WireGuard

home gateway
10.66.66.1/24

FORWARD + MASQUERADE

ONE ROUTE

bring up carrier

wg-quick up wg0

public IP = home

The P2P carrier reaches the home peer first; WireGuard then routes all Internet traffic through it.

The same need keeps repeating: reach a private service without owning its network edge.

Home lab
SSH · NAS
TCP

Family support
RDP · web UI
TCP

Remote work
PostgreSQL · dev API
TCP

Private browsing
SOCKS5h
SOCKS

Legacy VPN
OpenVPN
STREAM

Full tunnel
WireGuard
UDP

One stable PeerId replaces per-network IP addresses, port mappings, and temporary VPN control planes.

PeerId solves identity. The route still has to be found.

CLASSIC FORWARDING

The IP changes – access breaks

192.0.2.10:22

NAT / CGNAT

DNS, port forwarding, a VPN concentrator, or manually operated infrastructure.

P2P-NETCAT

Identity stays stable

12D3KooW…YXdUfbR9

mDNS · DHT · signaling · relay hints

PeerId selects the right peer; discovery and route racing select a reachable path.

The first route to complete authentication wins.

CLIENT

CLI or browser PWA
PeerId + logical port

PUBLIC ROUTING

mDNS · GossipSub · DHT

NATIVE SIGNALING

Nostr + WebTorrent

LIBP2P PLANE

QUIC · WebRTC Direct · WSS · TCP · Relay

PION WEBRTC

ICE/STUN · ordered DataChannel

AUTH GATE

Expected PeerId
admission handshake

LISTENER

Persistent Ed25519 identity

logical service
1…65535

SESSION AFTER AUTH

raw / exec

TCP forward

UDP frames

SOCKS

PTY / ConPTY

One CLI covers terminals, streams, datagrams, and proxies.

RAW / EXEC
stdin/stdout
stream or command
byte stream
-e

TCP FWD
127.0.0.1:PORT
dial TCP target
connection
-p

UDP FWD
127.0.0.1:PORT/udp
connected UDP socket
packet boundaries
-u -p

SOCKS
SOCKS4/4a/5
CONNECT remotely
remote DNS
-S

PTY
interactive TTY
Unix PTY / ConPTY
resize + EOF
-i

LISTENER
logical port
multiple sessions
identity + policy
-l -k

Wire protocol compatibility
/p2p-netcat/1.0.0/<port> · /p2p-netcat/udp/1.0.0/<port>

Application bytes flow only after the expected PeerId is proven.

Offer
Client creates SDP

DataChannel
p2p-netcat-v2

Challenge
32 random bytes

Signature
Ed25519 + service

PeerId check
Exact match

AUTH_READY
Session allowed

AFTER AUTH
256
KiB
end-to-end flow window

120 sec
reconnect on the same logical stream

UDP packet boundaries survive even over TCP/WSS/relay.

LOCAL SOURCE
127.0.0.1:15182
association per endpoint

00 84
uint16 BE length
exact UDP payload bytes

P2P STREAM
authenticated + encrypted
ordered reliable carrier

FIXED UDP TARGET
WireGuard · DNS · game service
exact reply to original source

Native WebRTC
ICE/STUN direct

QUIC / WebRTC Direct
direct UDP path

TCP / WSS
UDP-over-stream

Circuit Relay v2
fallback path

Tor + TCP relay
restricted networks

Fallback cost: ordered delivery can introduce head-of-line blocking.

The carrier stays outside the default route, so the tunnel cannot consume itself.

NAT A · CLIENT

WireGuard

Endpoint
127.0.0.1:15182

p2p-nc

UDP carrier
special UID

ip rule uidrange → lookup main

PUBLIC INTERNET

Nostr/WebTorrent signaling + STUN/ICE
application packets travel over selected P2P carrier

NAT B · GATEWAY

p2p-nc

UDP listener
→ UDP :51820

WireGuard

wg0 gateway
10.66.66.1/24

FORWARD + MASQUERADE

E2E CHECK

two NAT namespaces

WireGuard handshake

HTTP egress = full-tunnel-ok

Start the carrier first, then wg-quick up wg0 — policy routing protects DNS, signaling, STUN, and reconnect.

One primitive operation powers many working tools.

OpenSSH
ProxyCommand / :2222
TCP

PostgreSQL
:15432 → :5432
TCP

RDP
:13389 → :3389
TCP

SOCKS5h
remote DNS
SOCKS

OpenVPN
tcp-client
STREAM

WireGuard
full Internet tunnel
UDP

The logical port selects the P2P service; it does not have to match the local TCP/UDP port.

One installation command; four delivery surfaces.

VERIFIED DEPLOY

curl -fsSL …/deploy.sh | bash

detects OS + architecture

downloads the release

verifies SHA256SUMS

installs p2p-nc + pnc

RELEASE BINARIES

Linux · macOS · Windows · Android
amd64 / arm64

GHCR CONTAINER

ghcr.io/santaklouse/go-p2p-netcat
non-root · /config identity

GO INSTALL

cmd/p2p-nc + cmd/pnc
CGO is not required

BROWSER PWA

Static

GitHub Pages

React UI

Web Worker

IndexedDB routes

Native WebRTC

Tests target the exact failure boundaries introduced by the architecture.

NATIVE WEBRTC SOAK

60 / 60

workload iterations

1 152 MiB

transferredBytes in a fresh JSON report

Pion DataChannel · SHA-256 · reconnect

CI COVERAGE

Go tests

Linux · macOS · Windows

Race detector

Linux · ./...

Full tunnel

2 NAT + WireGuard + HTTP

Docker

buildx + runtime checks

WebRTC soak

weekly Ubuntu + macOS

WHAT SOAK DOES NOT PROVE

public Nostr/tracker availability

intercontinental latency

every NAT and firewall type

background browser tabs and sleep

100% relay-free connectivity

Three command pairs cover the highest-value use cases.

SSH THROUGH A LOCAL PORT

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 22 22022
client$ p2p-nc -p 2222
12D3KooWQ3uxpHgjDKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
22022
client$ ssh -p 2222 alice@127.0.0.1
```

TCP PORT FORWARDING

```
server$ p2p-nc -l -k -d 127.0.0.1 -p 5432 25432
client$ p2p-nc -p 15432
12D3KooWQ3uxpHgjDKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
25432
```

WIREGUARD CARRIER FOR A FULL TUNNEL

```
server$ p2p-nc -u -l -k -d 127.0.0.1 -p 51820 35182
client$ sudo wireguard-full-tunnel.sh --
/usr/local/bin/p2p-nc -u --udp-idle-timeout 0 \
          -p 15182
12D3KooWQ3uxpHgjDKE6vGmvzKS8RPbxUDLwJ7XCLaD6YXdUfbR9
35182
client$ sudo wg-quick up wg0
```

Use p2p-netcat when identity should outlive the address.

PeerId is the stable entry point. Route racing and fallback are the delivery mechanism.

github.com/santaklouse/go-p2p-netcat

GOOD FIT

access to home and edge hosts

SSH and private TCP services

UDP/WireGuard across NAT

temporary tunnels without a control plane

PLAN B REQUIRED

symmetric NAT or blocked UDP

guaranteed enterprise ingress

unsupported SOCKS UDP ASSOCIATE

TURN · Circuit Relay v2 · reachable TCP/WSS route